

Application of Machine Learning to Fischer–Tropsch Synthesis for Cobalt Catalysts

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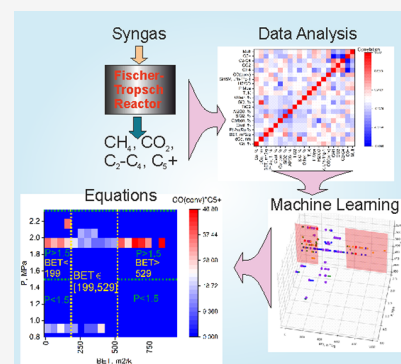


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ABSTRACT: Machine Learning was used to make a prediction model for six property parameters (COconv (%); CH₄ (%); CO₂ (%); C₂–C₄ (%); C₅+ (%); and COconv*C₅+ (%)) from 16 feature parameters of 169 Fischer–Tropsch synthesis experiments for cobalt catalyst. The Random Forest method was chosen as the “black-box” prediction tool, and cross-validation tests revealed small mean average errors: 10.9 ± 1.8 ; 3.7 ± 0.7 , 1.5 ± 0.6 , 3.6 ± 0.8 , 7.4 ± 1.4 and 9.0 ± 1.3 , respectively. The most important feature parameters were Brunauer–Emmett–Teller (BET) surface area, pressure (*P*), and temperature (*T*). The Decision Tree was chosen to build an explainable model, which revealed that BET in the range of [199.5, 529] leads to small COconv (%); C₅+ (%); and COconv*C₅+ (%) values. Outside this range, the samples tend to have large conversion and selectivity values. Most importantly, the BET effect is not linear or monotonic; it can be extracted using machine learning methods, and current work has shown how to use it. The resulting rules can be used for further experiments to maximize the efficiency and develop a new catalyst system. Particularly, porous Co₃O₄ with a small admixture of Al₂O₃ particles, Co₂O₃ by classical calcination of quartz sand impregnated with a cobalt nitrate solution, and a catalyst using a hydroxyapatite support were identified as good candidates for further deep screening of the best catalysts.



1. INTRODUCTION

Improving the efficiency of the deep processing of hydrocarbon raw materials is a highly relevant scientific and practical task in the expansion of the use of raw materials. The rate of growth of consumption of hydrocarbon products in the world steadily exceeds the dynamics of growth of demand for hydrocarbon raw materials.¹

One of the byproducts in the hydrocarbon production process for oil producers is associated petroleum gas (APG). APG is valuable for the production of synthetic liquid hydrocarbons from C₆ to C₂₀ and olefins.² What in the future can be used for complex petrochemical processing with obtaining high-octane components,³ solvents,⁴ aromatic hydrocarbons,⁵ alcohols,⁶ different inhibitors,⁷ synthetic fibers, and plastics,⁸ it underlines the high relevance of APG involvement in processing.

One of the areas of rational use of APG is the development of catalytic technology with low capital costs for efficient processing into transportable components of natural gas and commercial oil.⁹ This approach assumes conversion of part of the associated petroleum gas to produce hydrogen and synthesis gas, followed by conversion in the Fischer–Tropsch (FT) process.¹⁰

On the one hand, the approach to associated petroleum gas conversion based on the FT process has been known for quite a long time; on the other hand, until now, the applied

technologies are characterized by high capital and operating costs,¹¹ not high enough technological efficiency,¹² insufficient safety level,¹³ environmental problems¹⁴ (environmental pollution by hydrocarbons, greenhouse gas emissions—CO₂, CH₄), and decrease in obtaining valuable products with high added value.¹⁵ This is due to the fact that in traditional catalytic systems (granulated catalysts, monolithic ceramic blocks), it is difficult to implement effective management of thermal conditions of key reactions, which have high thermal effects and high sensitivity to temperature.¹⁶ Such catalytic systems are characterized by substantial diffusion inhibition of catalytic reactions and low thermal conductivity.¹⁷ The latter factor is a high difficulty in controlling the thermal conditions of the target reactions due to their high thermal effects, which leads to the need to use expensive and difficult-to-control reactors and heat exchanger bundling.¹⁸ As a result, traditional APG conversion technologies have low or even negative profitability.

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To overcome these problems, it is necessary to solve a fundamentally important scientific problem in the field of theoretical foundations of chemical technology: to develop a new catalytic system. Combining high catalyst activity, high intensity of mass transfer, low hydraulic resistance, and high effective thermal conductivity of catalytic cartridge. This combination provides high-quality thermal management. This opens the way to significant simplification and cheapening of catalytic reactors in this process.

One promising solution to this problem is the use of machine learning to find, develop, and optimize a new generation of catalytic systems. Such systems should have the highest ratio of mass transfer efficiency to pressure drop among all known catalytic structures while ensuring a high thermal conductivity of the catalyst. Catalysis informatics is a new subfield that combines cheminformatics and materials informatics to address the unique challenges of heterogeneous catalysis, utilizing data infrastructure, statistics, machine learning, and computational methods to generate explanatory and predictive hypotheses.^{19,20} Despite the hindrance of negative results in data science approaches, these findings can serve as valuable resources to evaluate the challenges in data-driven catalysis research and offer a roadmap for progress.²¹

To this day, a huge number of results of scientific experiments from all over the world are available in terms of conversion of synthesis gas to FT. The main parameters of synthetic product yield have been known for a long time: temperature, catalyst composition, pressure, and feed rate of reaction mixture into the reactor.²² However, the contribution of the influence of various factors on the level of CO conversion and the yield of the product as a whole is not completely clear. Although there are obvious trends.^{23–25} To systematize and identify patterns, the best solution is to use various machine learning (ML) methods. Thus, already for FTS, a multigressive ML method was used to determine the conditions of the best yield of light hydrocarbon fractions and, in particular, light alpha-olefins. For this method, the synthetic data calculated by the Single-Event MicroKinetic method were taken in a larger amount (2450), which deduced the rule and the main factors in the form of the temperature factor.²⁶ Methods hybrid GA-Fuzzy model, an artificial neural network model, and a comprehensive kinetic model were developed to establish the main rules for empirical data. Using a small amount of experimental data (30), the main rules were derived, but the data do not have a large scatter, and the optimal values for the conversion of CO and yield of C5+ are not so large.²⁷

2. EXPERIMENTAL SECTION

2.1. Catalyst Preparation. Comerica silica fiber SiO₂ (KT-11-TO, NPO Stekloplastik JSC, Russia) was used as a catalyst carrier with a specific surface area of 0.9 m² g⁻¹. Co/SiO₂ was prepared by the incipient wetness impregnation technique. The mass loading of Co was taken at the level of 15 wt % in relation to the sum of Co₃O₄ and SiO₂ carrier. Ten grams of silica fiber were impregnated with an aqueous solution (6 mL) of 5.49 g Co(NO₃)₂•6H₂O (pure, Khimprom-M LLC, Russia). After impregnation, the sample was dried at room temperature for 12 h, then dried at 100 °C for 1 h, and finally, the catalyst was calcined at 350 °C for 4 h. One sample was obtained for verification.

2.2. Catalyst Characterization. The surface area and pore structure of all the samples were determined by N₂ physical

adsorption–desorption at –196 °C, using the multipoint BET analysis method with a Micromeritics ASAP 2020 Plus instrument. Prior to the measurements, the samples were degassed at 250 °C for 6 h. The BET for the prepared sample is 40 m² g⁻¹.

The powder diffraction data for Rietveld analysis were collected at room temperature with a Tong-da t-3500 (China) powder diffractometer with Cu K α radiation and a linear detector. The step size of 2 θ was 0.016°, and the counting time was 0.2 s per step. All peaks were indexed by cubic cell (*Fd*-3*m*) with parameters *a* = 8.093(13) Å, *V* = 530(3) Å³ close to Co₃O₄. Therefore, this structure was taken as a starting model for Rietveld refinement, which was performed using TOPAS 4.2. Refinements were stable and gave low *R*-factors: *R*_{wp} = 5.87%; *R*_p = 4.21%; χ^2 = 1.95, *R*_B = 0.81% (Figure 1). During

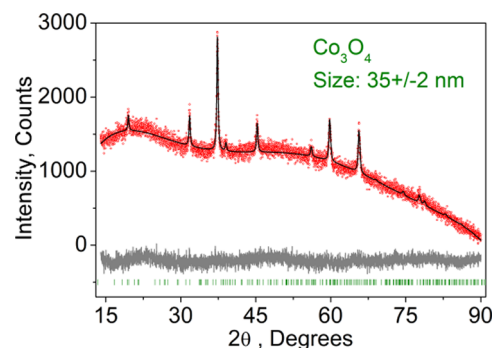


Figure 1. Difference Rietveld plot Co₃O₄ catalyst sample for test.

the refinement process, the average crystal size was estimated to be approximately 35 ± 2 nm. This crucial parameter *d*Co was subsequently utilized in our Machine Learning model to accurately predict properties.

2.3. Catalytic Evaluation. The Fischer–Tropsch synthesis was carried out in a fixed bed reactor at 12.5 bar, 250 °C, H₂/CO = 2 V/V and GHSV = 2.3 L g⁻¹ h⁻¹ using 2.85 g of catalyst. Before the reaction, the catalyst was reduced in situ with pure H₂ at 400 °C at a gas flow rate of 5 N L/h for 3 h. Then, the pressure was gradually increased to 13 bar, and syngas (H₂/CO = 2, V/V) was introduced at a gas flow rate of 6.6 N L/h.

Gaseous products were analyzed online on a Chromatec-Crystal 5000.2 gas chromatograph using a carbon molecular sieve column (NaX 60/80, Haysep R 80/100) with a thermal conductivity detector (TCD) using helium as a carrier gas. Liquid products collected from the cold trap were analyzed offline by GC–MS. Hydrocarbon products were extracted from the liquid product mixture with carbon tetrachloride. The extract was analyzed on an Agilent 5977B GC/MSD instrument on an Ultra ALLOY+5 capillary column with helium as a carrier gas.

3. SAMPLING METHOD

In this work, it was decided to establish with the method of machine learning what parameters of FTS and composition of the catalyst itself will be optimal for the maximum yield of heavy hydrocarbon fractions. To analyze the experimental data of the published works, selection criteria were established.

The Fischer–Tropsch synthesis on a cobalt catalytic system at a H₂/CO ratio (2 to 1) was chosen as a model reaction in order to obtain the C5+ hydrocarbon fraction. The choice of

the end product is driven by the widespread use of paraffins as lubricants for rubbing metal parts to increase their service life (the lubricant market was 39.71 billion liters at the end of 2021 and is projected to grow at an average annual rate of 3.52% to reach 47.22 billion liters in 2026).²⁸

The main selection parameter was the condition that FT synthesis occurs in a microchannel reactor with a fixed layer. The selection of scientific studies was made on the basis of the context and as much as possible of the performed work. As a result, 169 FTS examples of different compositions of cobalt-based catalysts under different synthesis conditions from 31 studies were used.

The following variation parameters were used for the model:

Temperature parameter.⁴ The choice is conditioned based on the Arrhenius equation. As can be seen in eq 1, the temperature is under the exponent, which affects the rate of the catalytic reaction.

The pressure parameter created in the system.²⁹ If we rely on the experimental data⁴ and on the given formula Arrhenius (eq 2), it is obvious that the rate of the reaction also depends on the pressure.

$$r = A \cdot \exp\left(-\frac{E_a}{RT}\right) \quad (1)$$

$$r_i = A_i \cdot \exp\left(-\frac{E_i}{RT}\right) \cdot P_{\text{CO}}^{m_i} P_{\text{H}_2}^{n_i} \quad (2)$$

The parameters of the volumetric gas flow rate (GHSV) and the ratio of the synthesis gas fed (H_2/CO). GHSV by definition is equal to the gas feed rate divided by the catalyst volume, but in some sources, the information on gas flow per catalyst mass was given.³⁰ In order to bring everything to one value, it was accepted to consider the volumetric gas flow rate as a ratio of the gas flow rate to catalyst mass. The conversion was performed by calculating the density of the fluidized bed. The higher the gas velocity that is fed into the reactor, the greater the conversion can be, but there are also limits above which the conversion will stop growing, and sometimes it can even drop. This limit depends on system parameters and reaction rates.³¹ Therefore, this parameter cannot be excluded. The H_2/CO ratio in some of the selected articles was different, which can affect the selectivity of the products.

The Brunauer–Emmett–Teller (BET) surface area parameter has a dimension of m^2/g . This parameter is a multipoint measurement of the specific surface area at which the synthesis gas is continuously passed through the catalyst. Its value plays a major role in the efficiency of the catalytic reaction, because the greater the surface area of the catalyst through which the synthesis gas passes, the greater will be the CO selectivity and hydrocarbon conversion.³²

The catalyst mass parameter is one of the obvious parameters that affect the CO conversion yields. Even if we look at the GHSV dimension, which, by definition, refers to the volume or mass of the catalyst, the mass of the catalyst can affect the product yield. To tame the calculation, we neglected the morphology of the catalyst particles. The catalyst particle sizes were taken into account based on XRD calculations.

The parameters of catalyst composition are a complex value. The parameter was divided into subparameters, the mass fraction of the catalyst composition. The following subparameters were distinguished: the fraction of cobalt as the main catalytic component; precious metals were singled out

separately to assess their necessity; zeolites, silicon oxides, aluminum oxide, titanium oxide, and carbon which are used as substrates for the catalyst itself. All of the above elements are the most popular carriers that increase the reaction surface area.³³ All other additives were combined into one parameter (other): zirconium,^{34–36} manganese,³⁷ sodium,³⁸ magnesium oxide,³⁹ phosphorus,⁴⁰ metallic aluminum,⁴¹ bismuth,⁴² hydroxyapatite,⁴³ iron,⁴⁴ cerium,⁴⁵ and cadmium.⁴⁶

The economic benefit of the relative cost of the catalyst and lifetime was not evaluated since the price of the carriers and cobalt salts themselves can vary greatly depending on the region and the purpose of the Fischer–Tropsch synthesis. However, the influence of rare elements (Ru, Pt, Au) on the Fischer–Tropsch synthesis was taken into account.

4. DATA AND METHOD

4.1. Data set Description. The data set of 169 experiments with 16 feature parameters (Co (%); dCo (nm), BET (m^2/g), Pt, A, Ru (%); Ceol (%), carbon (%), SiO_2 (%), Al_2O_3 (%); TiO_2 (%); SiC (%), Other (%); T (K), P (MPa); H_2/CO , GHSV ($\text{L h}^{-1} \text{g}^{-1}$); mcat (g))) and 5 property parameters (COconv (%); CH_4 (%); CO_2 (%); C2–C4 (%); C5+ (%)) was chosen for analysis. The descriptive statistics and the plot of the considered data set are summarized in Table S1 and Figure S1, respectively. Our aim was to find the experimental path, i.e., a combination of feature parameters to obtain the maximum COconv and C5+ simultaneously, i.e., we have chosen Mult = $\text{COconv} \cdot \text{C5+}$ as the goal property parameter. The maximum of the Mult $\sim 68\%$ is our goal, and experimental conditions that lead to it should be highlighted.

Preliminary data analysis revealed that Mult values showed an almost uniform distribution (Figure S1), which means that all representative experiments were selected. The correlation matrix (Figure 2) proves the absence of linear relationships between features and properties, which justifies the need to use Machine Learning methods. This matrix also revealed negative correlation between C5+ and CH_4 , CO_2 , C2–C4 which is expected because sum of all products is constant. One feature–feature correlation was detected between SiC and dCo values,

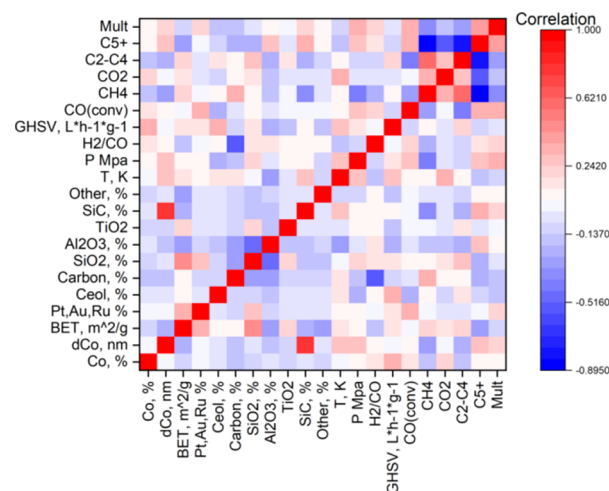


Figure 2. Correlation matrix of all feature, properties parameters, and property Mult = $\text{COconv} \cdot \text{C5+}$ revealed no valuable correlations between properties and features meaning that linear regression analysis will not reveal feature $\leftrightarrow$ property relationships.

which is actually accident and does not influence on model building.

4.2. Random Forest Method. The prediction tool used in this study is random forest (RF), an ensemble method based on regression trees.⁴⁷ The regression trees are built by recursive binary partitioning of the multidimensional predictor space into regions by constructing a multitude of “decision trees” at training time and outputting the class that is the “mode” of the classes (classification) or mean/average prediction (regression) of the individual trees.⁴⁸ Predictions are done by passing new data parameters from the root through the internal nodes until a terminal node is reached.

To start, the data set is divided into two distinct groups: the training data set and the test data set. The training data set usually comprises 70–90% of the total data set, while the test data set makes up the remaining 10–30%. The purpose of the training data set is to randomly select relevant features and samples in order to construct the Decision Tree model (Figure 3). This process is repeated until an optimal ensemble of N

total data). The Mean Absolute Error (MAE) and R^2 -values of the training set and test set and 5-fold cross-validation tests for all predicted values are summarized in Figure 4. We can conclude that the main correlation between experimental features and properties was revealed (Figure 2) which really demands using ML methods. The calculated Mult values and their differences from observed ones were less than 9% (Figure 4f), and a good fit between them can be checked in Figure 4f. The RF model can now be used to predict Mult values for any combinations of feature experimental parameters. It is important to note that a few data points may significantly deviate from commonly observed patterns or rules. Such deviations can be attributed to various factors, including experiment conditions that are not easily replicated, experimental errors, or unique circumstances that do not align with the majority of the samples. We recognize and appreciate the significance of these cases, and we see immense value in delving deeper into them for future exploration.

The measured feature parameters Co = 15%; dCo = 35(2) nm; BET = 40 m²/g; Pt, Au, Ru = 0%; Ceol = 0%; Carbon = 0%; SiO₂ = 85%; Al₂O₃ = 0%; TiO₂ = 0%; SiC = 0%; Other = 0%; $T = 523$ K; $P = 1.25$ MPa; $H_2/CO = 2$, GHSV = 2.32 L/h/g; $m(\text{cat}) = 2.85$ g, were used to accurately predict all six properties (COconv (%); CH₄ (%); CO₂ (%); C2–C4 (%); C5+ (%), Mult (%)) of the Co₃O₄ sample in order to conduct model comparing with ground truth experiments. The RF model evolved such values: COconv = $43 \pm 10\%$; CH₄ = $17 \pm 4\%$; CO₂ = $10 \pm 2\%$; C2–C4 = $14 \pm 4\%$; C5+ = $63 \pm 7\%$; Mult = $27 \pm 9\%$. The measured properties were as follows: COconv of 28%, CH₄ of 13%, CO₂ of 8%, C2–C4 of 2.1%, C5+ of 76.9%, and Mult = 21.5%. It is worth noting that all of these values, except for C2–C4, are closely aligned with the predicted values, falling within a range of 1–2 esd. The C2–C4 value aligned with predicted values within a range of 3 esd.

It should be noted that RF allows us to measure the importance of the feature after training. The selected value is permuted among the training data, and the error is computed on this perturbed data set. The importance score for the selected feature is computed by averaging the difference in error before and after the permutation over all trees.⁵⁰ The score is normalized by the standard deviation of these differences. Features that produce large values for this score are ranked as more important than features that produce small values. All importance parameters for predicted values are presented in Figure 4. The BET, P, and T feature parameters can be selected as the most important parameters to predict all properties. It is important to mention that SiO₂ is a highly significant parameter in predicting C2–C4. Similarly, the Co parameter plays a crucial role in forecasting CO₂, while dCo significantly affects COconv and GHSV to predict Mult = CO*C5+.

4.3. Decision Tree Method. In the previous section, we have got prediction RF “black box” and important feature parameters, which have the greatest influence on Mult value, but our general aim is to find the best rules which will lead to the highest Mult, i.e., interpretation of the prediction, so-called “white box”. It is well-known that the Decision Tree (DT) is one of the best methods to get interpretation.^{47,51}

The process of building a Decision Tree starts by sorting the data set according to a specific feature parameter, referred to as i . Then, the properties are divided into two subsets, Y1 and Y2, based on a value of m_i (as illustrated in Figure 5). The objective is to minimize the sum of dispersions (D) or

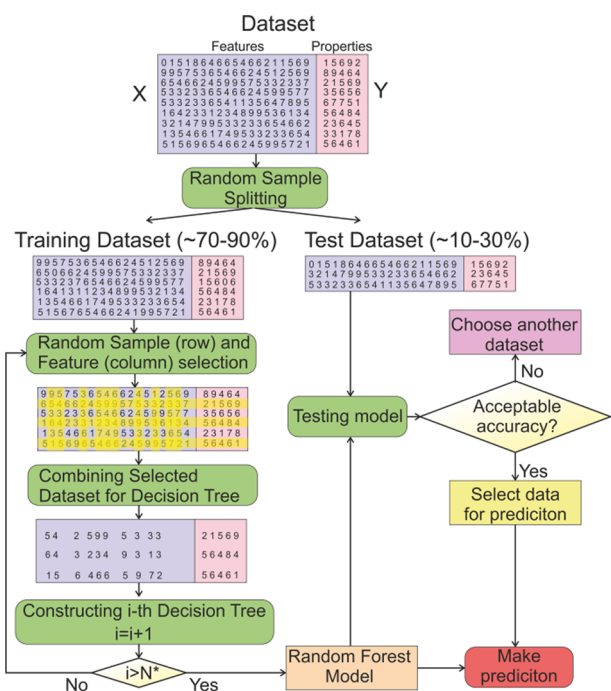


Figure 3. Main sequence of tasks comprises data set processing as well as building, validating, and generating predictions using the Random Forest model.

trees is created, where N is the only essential hyperparameter that needs to be fine-tuned in the Random Forest method. Once the ensemble of trees is formed, it is evaluated using the test data set. In the case that the model achieves a low mean absolute error (MAE) or high accuracy, it is considered reliable and well-suited for making predictions.

We used a simple self-written Python script named RandomForest.py using the Python 3.6 programming language⁴⁹ in order to build the described RF model. The standard libraries were used in the program: numpy, pandas, sklearn, matplotlib, and mpl_toolkits. Since this machine-learning algorithm is stochastic, we used it to average performance across ten repeats of cross-validation. Each time the data were split into two random data sets: a set for training procedure (80% of total data) and another set for test (20% of

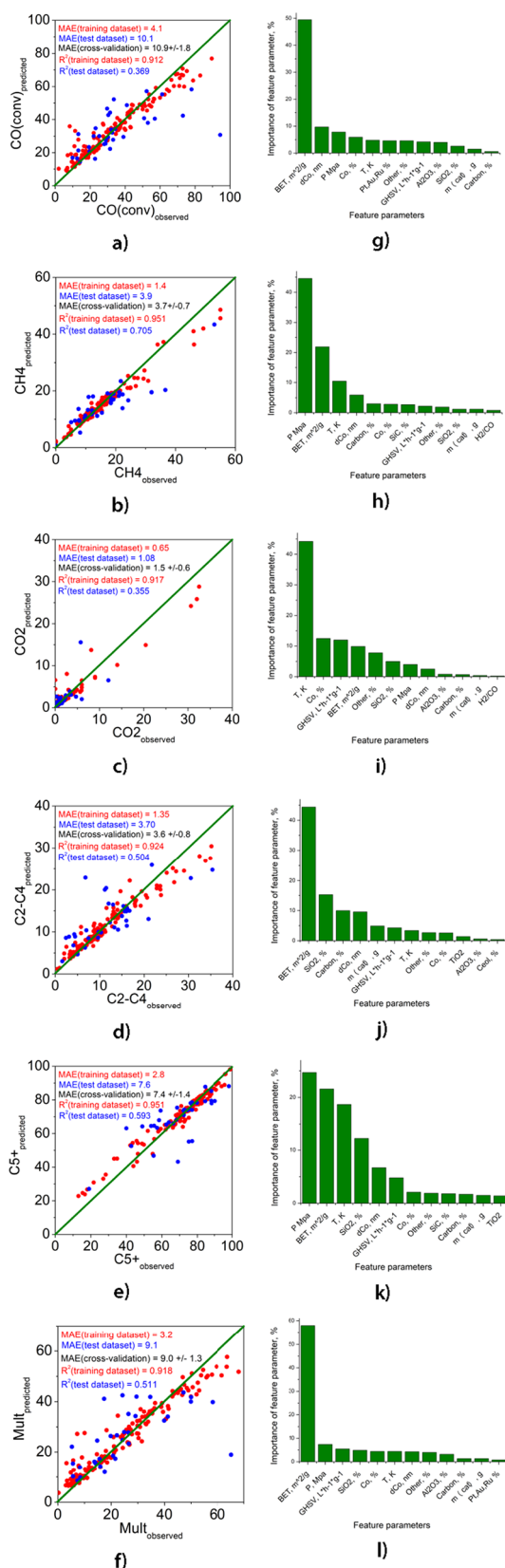


Figure 4. Comparative plot of observed values per calculated values obtained from RF model from (a) COconv; (b) CH₄; (c) CO₂; (d) C₂–C₄; (e) C₅⁺; (f) Mult = CO₂*C₅⁺. Linear fit proves correctness of models. (b) Importance of all feature parameters in RF model for: (g) COconv; (h) CH₄; (i) CO₂; (j) C₂–C₄; (k) C₅⁺; (l) Mult = CO₂*C₅⁺. One of the most important parameters is BET and P, T.

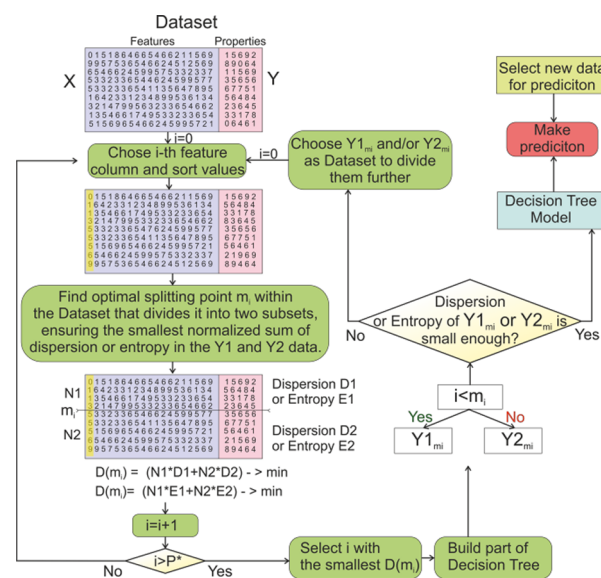


Figure 5. Flowchart outlines a step-by-step process consisting of three main components: data set processing, development of a Decision Tree model, and the generation of accurate predictions using this model.

entropies (*E*) of these subsets. This calculation is done by multiplying the number of data points in each subset (*N*₁ and *N*₂) with their corresponding dispersion (*D*₁ and *D*₂) or entropy (*E*₁ and *E*₂) values. This procedure is repeated for all feature parameters, and the parameter with the lowest *D* or *E* value is chosen. By applying the rule $i < m_i$, the upper section of the Decision Tree is constructed. The subsets *Y*₁ and *Y*₂ can be further divided using the same approach until the entire Decision Tree is formed. Once the tree is ready, it can be utilized for outcome prediction. This is achieved by passing new data parameters through the root node and then navigating through the internal nodes until reaching the terminal node, which provides the final prediction.

We used a simple self-written Python script named *DecisionTree.py* using the Python 3.6 programming language⁵⁰ in order to build the DT model. The standard libraries were used in the program: numpy, pandas, sklearn, matplotlib, and mpl_toolkits. In order to simplify the task, the DT with a depth equal to 3 was used. One of the main rules is $BET < 199.5$ which segregates 84 samples with large Mult = 31.6 ± 16 from 85 samples with low Mult = 19.5 ± 14 (Figure 6). An additional rule to this one is $P > 1.5$ MPa, which excludes 6 bad samples from this set. The second rule, which

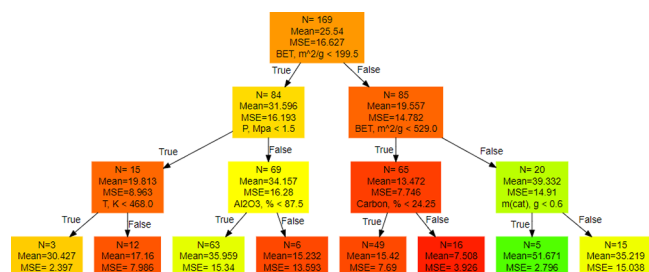


Figure 6. Decision Tree structure obtained from whole data set of 169 experiments. All bars which depict root, nodes, or leaves of the tree have colors from green (the highest Mult) to red (the smallest Mult) though yellow (average Mult).

can be highlighted is $BET > 529$ which segregates 20 samples with $Mult = 39 \pm 14$ from a relatively bad data set of 85 samples with $Mult = 19 \pm 14$ (Figure 6). Also, it should be noted that BET should not be bigger than 986 and P should not be bigger than 2.4 MPa, because collected samples have such limits. Therefore, we can summarize the most optimal feature parameter intervals for high Mult values: $BET \in [0, 199.5]$ V $[529, 986]$ and $P \in [1.5, 2.4]$.

The data plot of COconv, C5+, and Mult values in the space of main feature parameters (BET, P, T) is depicted in Figure 7.

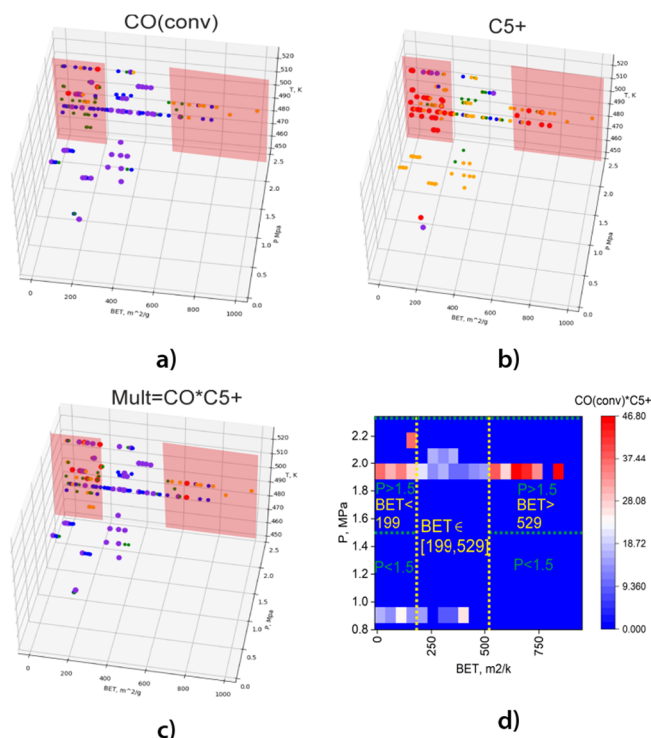


Figure 7. Collected experimental data plot in the most important feature (BET, P, T) parameter space: (a) COconv; (b) C5+; (c) $Mult = COconv \cdot C5+$. Red dots depict 80–100% of maximum property values, orange dots –60 to 80%, green –40 to 60%, blue –20 to 40%, and violet –0 to 20%. The zones with high COconv, C5+ and Mult values are highlighted. Pictures prove selected rules for high Mult values: $BET \in [0, 199.5]$ V $[529, 986]$ and $P \in [1.5, 2.4]$. Averaged $CO(conv) \cdot C5+$ plot of experimental data in 2D (BET, P) space, zones with high and small values of conversion, and C5+ selective values are segregated by equations.

One can see that highlighted zones of select rules $BET \in [0, 199.5]$ V $[529, 986]$ and $P \in [1.5, 2.4]$ really segregates high property values (red, orange dots) from low values (blue, violet). Therefore, one can follow these simple rules to get both high C5+ and CO(conv). In order to get the most precise value, of course, it will be better to use the Random Forest model, described above.

The H_2/CO ratio is not the decisive factor in the decision tree. However, increasing the H_2/CO ratio leads to faster deactivation of the cobalt catalysts. This is due to the reduced power of hydrogen. At a lower H_2/CO ratio, sintering of the catalyst occurs. Given that the catalyst showed the lowest degree of deactivation at an H_2/CO ratio of 1. We can conclude that sintering was less important for catalyst deactivation.⁵²

Three zones are clearly distinguished: two zones with effective catalysts and an intermediate zone with poorly effective catalysts. On the one hand, the zone with $BET \in (199.5, 529)$ is difficult to separate in composition from the other zones; on the other hand, catalysts with the zeolite substrate are clearly distinguished. All catalysts of this type show efficiencies ($CO \cdot C5+/100$) of less than 15%, except for one sample with the lowest zeolite content (20%) and an efficiency of 30%.⁵³

Among the remaining two areas, it is difficult to identify any particular composition or condition. However, five samples with efficiencies >60% stand out. Two of these samples consist mainly of porous Co_3O_4 with a small admixture of Al_2O_3 , 5 and 10%.⁵⁴ The efficiencies are 61.6 and 63.6%, respectively. A special feature of both samples is the preparation of the catalyst. Annealing in air with the addition of Al_2O_3 is carried out at 550 °C, which is due to which the compound $CoAl_2O_4$ is formed. According to the authors of the work, this allows to preserve the mesoporous structure of Co_3O_4 due to the formation of $CoAl_2O_4$ on the outer surface with less blocking of the active centers of metallic cobalt-formed heavy coke precursors.⁵⁴ However, the high CO_2 yield and high Co content make such catalysts unattractive.

A catalyst based on mesoporous silicon oxide (MS) 81% and titanium oxide 9% has a similar efficiency of 63.7%.⁵⁵ This is the only sample of the top 5 with a $BET > 500$ g/cm². It differs in its branch with a slightly elevated temperature of 503 K. The same sample at 493 K has an efficiency of 57.8%. The same efficiency drops sharply when the SiO_2 or TiO_2 fraction is increased, while the cobalt fraction remains at 10%. The authors of the work explain this by the fact that the addition of TiO_2 can effectively modulate the interaction between cobalt and the carrier. However, also a large amount of TiO_2 introduced reduces the degree of reduction and characteristics of FTS catalysts.⁵⁵

A more interesting case is the conventional application of Co_2O_3 particles through the classical calcination of quartz sand impregnated with cobalt nitrate solution.³⁸ The efficiency of the catalyst is 65%. The specific surface area of the purchased SiO_2 was 200 m²/g. The sample had a fairly high proportion of cobalt in its composition (20%). When comparing with samples of the same composition, the sample is strongly knocked out, and when comparing, the efficiency of three other samples of the same composition is from 18 to 10.5% (Table S1). The main difference is the temperature—the samples with low efficiency are 30 degrees lower (523 and 493 K, respectively). The second is the use of mesoporous silica (SBA-15).³⁷ From the example of the last sample, it is obvious that the use of mesoporous silica without additives leads to low conversion. However, it is not completely clear what factors may also affect the efficiency of catalysts based on non-mesoporous SiO_2 .

The most interesting case will result in the work of Munirathinam.⁴³ The authors of the work obtained a catalyst efficiency of 68% using a substrate of hydroxyapatite. It is shown that reducing GHSV by half (from 1.54 to 0.77) increases the conversion of CO by half, and increasing the temperature by 10 °C increases the conversion by a third. Taken together, changing only two parameters of GHSV and FT temperature increased conversion by a factor of 4. At the same time, if we consider the work, which contains aluminum phosphate $AlPO_4$, its catalytic properties are not so impressive.⁴⁰ Although the addition of a small amount of

AlPO₄ up to 9% increases the conversion of CO. However, a large phosphate content already decreases the efficiency. This clearly indicates that the hydroxy group content has a positive effect on the efficiency of the catalyst. These researches give grounds to investigate phosphate substrates with hydroxyapatite structure and also heat resistant minerals with hydroxyl group in composition by type biotite and Illite.^{56,57}

5. CONCLUSIONS

Machine learning was performed for 169 experiments with Co-containing catalysts. The Random Forest model was able to predict COconv, CH₄, CO₂, C2–C4, C5+ and Mult = CO*C5+ with small average cross-validation errors: 10.9 ± 1.8; 3.7 ± 0.7, 1.5 ± 0.6, 3.6 ± 0.8, 7.4 ± 1.4, and 9.0 ± 1.3, respectively. The most important parameters influencing these characteristics were BET, P, and T, as expected. However, a decision tree was used to build the explanation model. It was found that the feature parameters COconv, C5+, and Mult = COconv*C5+ have two zones with high values, and mathematically these areas can be expressed as BET ∈ [0, 199.5] V [529, 986] and P ∈ [1.5, 2.4]. Surprisingly, almost all compounds with BET ∈ [199.5, 529] showed small values of CO(conv), C5+, and Mult = COconv*C5+. Analysis of the catalysts in this area has revealed zeolite-supported catalysts that indeed show a low efficiency of less than 15%. Meanwhile, the analysis of catalysts in BET not in this range [199.5, 529] revealed 5 samples with high conversion and selectivity: two of which consist mainly of porous Co₃O₄ with a small admixture of Al₂O₃; mesoporous silicon oxide 81% and titanium oxide 9%; Co₂O₃ particles by classical calcination of quartz sand impregnated with cobalt nitrate solution; and a catalyst using a hydroxyapatite support. All these compounds can be considered as good candidates for further in-depth screening of the best catalysts, especially taking into account derived rules BET ∈ [0, 199.5] V [529, 986] and P ∈ [1.5, 2.4].

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.iecr.3c03147>.

Data set of 169 experiments with 16 feature parameters and 6 property parameters (COconv (%); CH₄ (%); CO₂ (%); C2–C4 (%); C5+ (%); Mult); descriptive statistics and the plot of the considered data set; and program code for implementing Decision Tree and Random Forest in Python3 (PDF)

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■ ABBREVIATIONS

APG, associated petroleum gas; BET, Brunauer–Emmett–Teller; FT, Fischer–Tropsch; FTS, Fischer–Tropsch Synthesis; GHSV, volumetric gas flow rate; ML, machine learning; MAE, mean absolute error; RF, Random Forest; DT, decision tree; MS, mesoporous silicon.

■ REFERENCES

- (1) Dudin, M. N.; Frolova, E. E.; Protopopova, O. V.; Mamedov, A. A.; Odintsov, S. V. Study of Innovative Technologies in the Energy Industry: Nontraditional and Renewable Energy Sources. *Entrepreneurship and Sustainability Issues* **2019**, *6* (4), 1704–1713.
- (2) Dolinskii, S. E. Economically Attractive Technologies of Deep Conversion of Associated Petroleum Gas. *Russ J. Gen Chem.* **2011**, *81* (12), 2574–2593.
- (3) Arutyunov, V. S. Utilization of Associated Petroleum Gas via Small-Scale Power Generation. *Russ J. Gen Chem.* **2011**, *81* (12), 2557–2563.
- (4) Fernandes, R. A.; Jha, A. K.; Kumar, P. Recent Advances in Wacker Oxidation: From Conventional to Modern Variants and Applications. *Catalysis Science and Technology* **2020**, *10*, 7448–7470.
- (5) Ramirez, A.; Dutta Chowdhury, A.; Dokania, A.; Cnudde, P.; Caglayan, M.; Yarulina, I.; Abou-Hamad, E.; Gevers, L.; Ould-Chikh, S.; De Wispelaere, K.; Van Speybroeck, V.; Gascon, J. Effect of Zeolite

- Topology and Reactor Configuration on the Direct Conversion of CO₂ to Light Olefins and Aromatics. *ACS Catal.* **2019**, *9* (7), 6320–6334.
- (6) Yan, M.; Lo, J. C.; Edwards, J. T.; Baran, P. S. Radicals: Reactive Intermediates with Translational Potential. *J. Am. Chem. Soc.* **2016**, *138*, 12692–12714.
- (7) Saleh, T. A.; Ibrahim, M. A. Synthesis of Amyl Ester Grafted on Carbon-Nanopolymer Composite as an Inhibitor for Cleaner Shale Drilling. *Petroleum* **2022**, *8* (4), 529–537.
- (8) Dvoynikov, M.; Buslaev, G.; Kunshin, A.; Sidorov, D.; Kraslawski, A.; Budovskaya, M. Resources New Concepts of Hydrogen Production and Storage in Arctic Region. *Resources* **2021**, *10*, 1–18, DOI: 10.3390/resources100.
- (9) Arutyunov, V. S.; Savchenko, V. I.; Sedov, I. V.; Nikitin, A. V.; Troshin, K. Y.; Borisov, A. A.; Fokin, I. G.; Makaryan, I. A.; Strekova, L. N. New Potentialities for Utilization of Associated Petroleum Gases in Power Generation and Chemicals Production. *Eurasian Chemico-Technological Journal* **2017**, *19* (3), 265–271.
- (10) Mahmoudi, H.; Mahmoudi, M.; Doustdar, O.; Jahangiri, H.; Tsolakis, A.; Gu, S.; LechWyszynski, M. A Review of Fischer–Tropsch Synthesis Process, Mechanism, Surface Chemistry and Catalyst Formulation. *Biofuels Engineering* **2017**, *2* (1), 11–31.
- (11) Zang, G.; Sun, P.; Elgowainy, A. A.; Bafana, A.; Wang, M. Performance and Cost Analysis of Liquid Fuel Production from H₂ and CO₂ based on the Fischer–Tropsch Process. *J. CO₂ Util.* **2021**, *46*, No. 101459.
- (12) Jones, M. P.; Krexner, T.; Bismarck, A. Repurposing Fischer–Tropsch and Natural Gas as Bridging Technologies for the Energy Revolution. *Energy Conversion and Management* **2022**, *267*, No. 115882.
- (13) Hamad, N. A.; El-Halwagi, M. M.; Elbashir, N. O.; Mannan, S. M. Safety Assessment of Potential Supercritical Solvents for Fischer–Tropsch Synthesis. *J. Loss Prev Process Ind.* **2013**, *26* (3), 528–533.
- (14) Wang, D.; Gu, Y.; Chen, Q.; Tang, Z. Direct Conversion of Syngas to Alpha Olefins via Fischer–Tropsch Synthesis: Process Development and Comparative Techno-Economic-Environmental Analysis. *Energy* **2023**, 263.
- (15) Steynberg, A. P. Introduction to Fischer–Tropsch Technology. *Stud. Surf. Sci. Catal.* **2004**, *152*, 1–63.
- (16) Gholami, Z.; Tišler, Z.; Rubáš, V. Recent Advances in Fischer–Tropsch Synthesis Using Cobalt-Based Catalysts: A Review on Supports, Promoters, and Reactors. *Catal. Rev. Sci. Eng.* **2021**, *63* (3), 512–595.
- (17) Merino, D.; Sanz, O.; Montes, M. Effect of the Thermal Conductivity and Catalyst Layer Thickness on the Fischer–Tropsch Synthesis Selectivity Using Structured Catalysts. *Chemical Engineering Journal* **2017**, *327*, 1033–1042.
- (18) Fratalocchi, L.; Visconti, C. G.; Groppi, G.; Lietti, L.; Tronconi, E. Intensifying Heat Transfer in Fischer–Tropsch Tubular Reactors through the Adoption of Conductive Packed Foams. *Chemical Engineering Journal* **2018**, *349*, 829–837.
- (19) Medford, A. J.; Kunz, M. R.; Ewing, S. M.; Borders, T.; Fushimi, R. Extracting Knowledge from Data through Catalysis Informatics. *ACS Catal.* **2018**, *8* (8), 7403–7429.
- (20) Takahashi, K.; Takahashi, L.; Miyazato, I.; Fujima, J.; Tanaka, Y.; Uno, T.; Satoh, H.; Ohno, K.; Nishida, M.; Hirai, K.; Ohyama, J.; Nguyen, T. N.; Nishimura, S.; Taniike, T. The Rise of Catalyst Informatics: Towards Catalyst Genomics. *ChemCatChem* **2019**, *11*, 1146–1152.
- (21) Taniike, T.; Takahashi, K. The Value of Negative Results in Data-Driven Catalysis Research. *Nature Catalysis* **2023**, *6*, 108–111.
- (22) Loseva, N. I.; Niyazbakiev, I. I.; Silman, A. V. Fischer–Tropsch Synthesis Technology Evolution. In *IOP Conference Series: Materials Science and Engineering*; IOP Publishing Ltd, 2019; Vol. 663.
- (23) Barrios, A. J.; Gu, B.; Luo, Y.; Peron, D. V.; Chernavskii, P. A.; Virginie, M.; Wojcieszak, R.; Thybaut, J. W.; Ordonsky, V. V.; Khodakov, A. Y. Identification of Efficient Promoters and Selectivity Trends in High Temperature Fischer–Tropsch Synthesis over Supported Iron Catalysts. *Appl. Catal., B* **2020**, *273*, No. 119028.
- (24) Dalai, A. K.; Davis, B. H. Fischer–Tropsch Synthesis: A Review of Water Effects on the Performances of Unsupported and Supported Co Catalysts. *Applied Catalysis A: General* **2008**, *348*, 1–15.
- (25) Seyednejadian, S.; Rauch, R.; Bensaid, S.; Hofbauer, H.; Weber, G.; Saracco, G. Power to Fuels: Dynamic Modeling of a Slurry Bubble Column Reactor in Lab-Scale for Fischer–Tropsch Synthesis under Variable Load of Synthesis Gas. *Applied Sciences (Switzerland)* **2018**, *8* (4), 514.
- (26) Chakkingal, A.; Janssens, P.; Poissonnier, J.; Virginie, M.; Khodakov, A. Y.; Thybaut, J. W. Multi-Output Machine Learning Models for Kinetic Data Evaluation: A Fischer–Tropsch Synthesis Case Study. *Chem. Eng. J.* **2022**, *446*, No. 137186.
- (27) Eghbal Ahmadi, M. H.; Mosayebi, A. Fischer – Tropsch Synthesis over Co-Ni/Al₂O₃ Catalyst: Comparison between Comprehensive Kinetic Modeling, Artificial Neural Network, and a Novel Hybrid GA-Fuzzy Models. *J. Taiwan Inst Chem. Eng.* **2021**, *127*, 32–45.
- (28) Jacobs, G.; et al. Fischer–Tropsch synthesis: support, loading, and promoter effects on the reducibility of cobalt catalysts. *Applied Catalysis A: General* **2002**, *233*, 263–281.
- (29) Sun, Y.; Jia, Z.; Yang, G.; Zhang, L.; Sun, Z. Fischer–Tropsch Synthesis Using Iron Based Catalyst in a Microchannel Reactor: Performance Evaluation and Kinetic Modeling. *Int. J. Hydrogen Energy* **2017**, *42* (49), 29222–29235.
- (30) Subramanian, V.; Zholobenko, V. L.; Cheng, K.; Lancelot, C.; Heyte, S.; Thuriot, J.; Paul, S.; Ordonsky, V. V.; Khodakov, A. Y. The Role of Steric Effects and Acidity in the Direct Synthesis of Iso-Paraffins from Syngas on Cobalt Zeolite Catalysts. *ChemCatChem* **2016**, *8* (2), 380–389.
- (31) Wang, X.; Ren, Y.; Zhang, L. Intensification of High-Flux Fischer–Tropsch Synthesis with Axial Gradient Co-Loading FeCrAl Monolith Catalysts. *Chem. Eng. J.* **2023**, *452*, No. 139188.
- (32) Lualdi, M.; Lögdberg, S.; Boutonnet, M.; Järås, S. On the Effect of Water on the Fischer–Tropsch Rate over a Co-Based Catalyst: The Influence of the H₂/CO Ratio. *Catal. Today* **2013**, *214*, 25–29.
- (33) Jacobs, G.; Das, T. K.; Zhang, Y.; Li, J.; Racollet, G.; Davis, B. H. Fischer–Tropsch Synthesis: Support, Loading, and Promoter Effects on the Reducibility of Cobalt Catalysts. *Applied catalysis* **2002**, *233*, 263–281.
- (34) Yaghoobpour, E.; Zamani, Y.; Zarrinpashne, S.; Zamaniyan, A. Fischer–Tropsch Synthesis: Effect of Silica on Hydrocarbon Production over Cobalt-Based Catalysts. *Chemical Papers* **2019**, *73* (1), 205–214.
- (35) Han, Y.; Xiao, G.; Chen, M.; Chen, S.; Zhao, F.; Zhang, Y.; Li, J.; Hong, J. Effect of Support Modification and Precursor Decomposition Method on the Properties of CoPt/ZrO₂ Fischer–Tropsch Catalysts. *Catal. Today* **2021**, *375*, 1–9.
- (36) Mohammadnasabomran, S.; Tavasoli, A.; Zamani, Y.; Márquez-Álvarez, C.; Pérez-Pariente, J.; Martínez, A. Influence of Amines in the Synthesis and Properties of Mesoporous Aluminas and Assessment as Supports for Co-Based Fischer–Tropsch Synthesis Catalysts. *Fuel Process. Technol.* **2020**, *205*, No. 106433.
- (37) Martínez, A.; López, C.; Márquez, F.; Díaz, I. Fischer–Tropsch Synthesis of Hydrocarbons over Mesoporous Co/SBA-15 Catalysts: The Influence of Metal Loading, Cobalt Precursor, and Promoters. *J. Catal.* **2003**, *220* (2), 486–499.
- (38) Dai, Y.; Yu, F.; Li, Z.; An, Y.; Lin, T.; Yang, Y.; Zhong, L.; Wang, H.; Sun, Y. Effect of Sodium on the Structure-Performance Relationship of Co/SiO₂ for Fischer–Tropsch Synthesis. *Chin. J. Chem.* **2017**, *35* (6), 918–926.
- (39) Zhang, Y.; Xiong, H.; Liew, K.; Li, J. Effect of Magnesia on Alumina-Supported Cobalt Fischer–Tropsch Synthesis Catalysts. *J. Mol. Catal. A Chem.* **2005**, *237* (1–2), 172–181.
- (40) Park, S. J.; Cho, J. M.; Ahn, C. Il; Lee, Y. J.; Jun, K. W.; Cho, B. G.; Bae, J. W. Roles of Phosphorous-Modified Al₂O₃ for an Enhanced Stability of Co/Al₂O₃ for CO Hydrogenation to Hydrocarbons. *J. Mol. Catal. A Chem.* **2017**, *426*, 177–189.
- (41) Wang, D.; Chen, L.; Li, G.; Wang, Z.; Li, X.; Hou, B. Cobalt-Based Fischer–Tropsch Synthesis: Effect of the Catalyst Granule

Thermal Conductivity on the Catalytic Performance. *Mol. Catal.* **2021**, 502, No. 111395.

(42) Gu, B.; Peron, D. V.; Barrios, A. J.; Virginie, M.; La Fontaine, C.; Briois, V.; Vorokhta, M.; Šmíd, B.; Moldovan, S.; Koneti, S.; Gambu, T. G.; Saeys, M.; Ordonsky, V. V.; Khodakov, A. Y. Bismuth Mobile Promoter and Cobalt-Bismuth Nanoparticles in Carbon Nanotube Supported Fischer–Tropsch Catalysts with Enhanced Stability. *J. Catal.* **2021**, 401, 102–114.

(43) Munirathinam, R.; Pham Minh, D.; Nzihou, A. Hydroxyapatite as a New Support Material for Cobalt-Based Catalysts in Fischer–Tropsch Synthesis. *Int. J. Hydrogen Energy* **2020**, 45 (36), 18440–18451.

(44) Griboval-Constant, A.; Butel, A.; Ordonsky, V. V.; Chernavskii, P. A.; Khodakov, A. Y. Cobalt and Iron Species in Alumina Supported Bimetallic Catalysts for Fischer–Tropsch Reaction. *Appl. Catal. A Gen* **2014**, 481, 116–126.

(45) Mohammadnasabomran, S.; Tavasoli, A.; Zamani, Y.; Márquez-Álvarez, C.; Pérez-Pariente, J.; Martínez, A. Influence of Amines in the Synthesis and Properties of Mesostructured Aluminas and Assessment as Supports for Co-Based Fischer–Tropsch Synthesis Catalysts. *Fuel Process. Technol.* **2020**, 205, No. 106433.

(46) Ma, W.; Jacobs, G.; Qian, D.; Ji, Y.; Klettlinger, J. L. S.; Hopps, S. D.; Davis, B. H. Fischer–Tropsch Synthesis: Synergistic Effect of Hybrid Pt–Cd Additives on a 15%Co/Al₂O₃ Catalyst. *Appl. Catal. A Gen* **2020**, 600.

(47) Breiman, L. Random Forests. *Machine Learning* **2001**, 45, 5–32.

(48) Kam Ho, T. K. Random Decision Forests. *Proceeding of 3rd international conference on document analysis and recognition* **1995**, 1, 278–282.

(49) Coelho, L. P.; Richert, W.; Brucher, M. *Building Machine Learning Systems with Python: Explore Machine Learning and Deep Learning Techniques for Building Intelligent Systems Using Scikit-Learn and TensorFlow*, 3rd ed.; Packt Publishing: Birmingham, 2018.

(50) Zhu, R.; Zeng, D.; Kosorok, M. R. Reinforcement Learning Trees. *J. Am. Stat. Assoc.* **2015**, 110 (512), 1770–1784.

(51) Custode, L. L.; Iacca, G. Evolutionary Learning of Interpretable Decision Trees. *IEEE Access* **2023**, 11, 6169–6184.

(52) Zhou, W.; Chen, J. G.; Fang, K. G.; Sun, Y. H. The Deactivation of Co/SiO₂ Catalyst for Fischer–Tropsch Synthesis at Different Ratios of H₂ to CO. *Fuel Process. Technol.* **2006**, 87 (7), 609–616.

(53) Liu, C.; Chen, Y.; Zhao, Y.; Lyu, S.; Wei, L.; Li, X.; Zhang, Y.; Li, J. Nano-ZSM-5-Supported Cobalt for the Production of Liquid Fuel in Fischer–Tropsch Synthesis: Effect of Preparation Method and Reaction Temperature. *Fuel* **2020**, 263, No. 116619.

(54) Koo, H. M.; Ahn, C. Il; Lee, D. H.; Roh, H. S.; Shin, C. H.; Kye, H.; Bae, J. W. Roles of Al₂O₃ Promoter for an Enhanced Structural Stability of Ordered-Mesoporous Co₃O₄ Catalyst during CO Hydrogenation to Hydrocarbons. *Fuel* **2018**, 225, 460–471.

(55) Wu, H.; Yang, Y.; Suo, H.; Qing, M.; Yan, L.; Wu, B.; Xu, J.; Xiang, H.; Li, Y. Effect of TiO₂ Promotion on the Structure and Performance of Silica-Supported Cobalt-Based Catalysts for Fischer–Tropsch Synthesis. *J. Mol. Catal. A Chem.* **2014**, 390, 52–62.

(56) He, C.; Makovicky, E.; Øsbæck, B. Thermal Stability and Pozzolanic Activity of Calcined Illite. *Appl. Clay Sci.* **1995**, 9, 337–354.

(57) Rouxhet, P. G.; Gillard, J. L.; Fripiat, J. J. Thermal Decomposition of Amosite, Crocidolite, and Biotite. *Mineralogical Magazine* **1972**, 38, 583–592.



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